



BSc (Hons) in **Information Technology**
Specialized in AI

IT3012 : Intelligent Agents

Year 3 · Semester 1 · 2026 Semester 2

Faculty of Computing

Practical 01

Objective & Lecture Mapping: Recall that an intelligent agent acts within a continuous loop: Sensors generate percept sequences, and Actuators execute actions to maximize a Performance measure. Use this sheet to execute practical modifications and incrementally evaluate your design against Lecture 01 and 02 theory (from basic recall to analytical classification).

Part 1: Code Analysis & PEAS Evaluation

Task 0 : Setting up and getting the starter Code

1. Go to the [Git Hub Repo](#). Create a Fork of this Git Repo to your Personal Github Profile. Open it using your favourite IDE and follow the below instruction to complete the practical. The base code is in the "master" brach of the repo.

Task 1.1: The Environment State (`__init__`)

1. (Remember) List the four components of the PEAS framework discussed in Lecture 01.

Performance Measure
Environment
Actuators
Sensors

2. (Understand) Look at the `VisualGridHuntGame.__init__` method. Which specific variables in this code represent the physical "Environment (E)" state?

The environment state is represented by variables that describe the world itself, such as

Grid dimensions

Wall locations

Food positions

Opponent positions



3. (Analyze) Based on the variables initialized (specifically `self.opponents`), classify this baseline environment as "Single-Agent" or "Multi-Agent". Briefly justify your choice.

Multi-Agent

The environment contains `self.opponents`, meaning other entities exist and can affect the environment and outcomes. Therefore, the agent is not acting alone.



Task 1.2: The Perception Subsystem (`get_percept`)

4. (Remember) Define what a "percept sequence" is according to Lecture 01.

A percept sequence is the complete history of all percepts received by an agent from the moment it starts until the current time.



5. (Understand) Which component of the PEAS framework does the `get_percept()` method represent in our code?

Sensors



6. (Evaluate) Based on the exact dictionary returned by `get_percept()` , is this environment "Fully Observable" or "Partially Observable"? Explain why based on what the agent can and cannot see.

Partially Observable

The agent only knows what is included in the percept returned by `get_percept()`. Any information not provided through the sensors is hidden from the agent. A partially observable environment occurs when some relevant state information is unavailable.

**Task 1.3: Action Execution (`execute_action`)**

7. (Understand) When `execute_action()` deducts points for hitting a wall, which PEAS component is being actively updated?

Performance Measure (P)

The score is being changed to evaluate how well the agent is performing.



8. (Evaluate) Why is it structurally important that this metric evaluates changes in the external environment state (hitting a wall) rather than the agent's internal processing effort?

Performance measures should evaluate the agent's effect on the environment rather than what happens internally. Measuring only internal activity can lead to exploitative behavior that maximizes score without accomplishing the real task.

**Part 2: Guided Modification & Deep Theory Mapping**

Follow the implementation instructions to inject a new hazard, then answer the questions to map your code changes back to the theoretical nature of the environment.

Step 2.1: Extending the Environment Initialization (`__init__`)

1. **Implementation Instructions:** Declare a new trap collection attribute (e.g., `self.toxic_traps = set()`) inside `__init__`. Populate it with coordinate tuples

safely avoiding position $(0, 0)$, walls, and food.

9. (Understand) If you successfully add `self.toxic_traps` to the environment but intentionally **hide** this data from the agent's sensors, how does this specifically alter the environment's "Observability" classification?

Hiding trap information makes the environment more partially observable, because the agent cannot directly perceive all relevant aspects of the world state.

Step 2.2: Updating the Perception Subsystem (`get_percept`)

1. **Implementation Instructions:** Locate `get_percept(self)` and add a new boolean sensor key (e.g., `'smells_toxin': tuple(self.agent_pos) in self.toxic_traps`) to the dictionary returned to the agent loop.

10. (Analyze) By adding `'smells_toxin'`, you expand the agent's percept sequence. Explain how this specific sensor helps the agent act with "Rationality" (maximizing expected utility).

It gives the agent additional information about potential danger. With better percepts, the agent can make decisions that maximize expected utility and avoid harmful actions. Rational decisions depend on the information available in the percept sequence

Step 2.3: Modifying Action Execution & Visual Rendering (`execute_action` & `draw_grid`)

1. **Implementation Instructions:** In `execute_action`, check if the updated `self.agent_pos` intersects with `self.toxic_traps` to decrement `self.score` by 15 points. In `draw_grid`, iterate through traps to render custom purple shapes on the canvas.

11. (Remember) You programmed a severe score penalty for stepping on a trap to avoid metric exploitation. What was the classic "Vacuum World Exploit" example discussed in Lecture 01 that illustrated the danger of faulty metrics?

A robot vacuum was rewarded whenever it collected dirt. A rational agent could exploit this by cleaning dirt, dumping it back onto the floor, and repeatedly cleaning the same spot to gain unlimited points. The lesson was that performance measures must reward the desired environmental outcome (clean rooms), not intermediate actions.



Finalize and Lock Lab 01 Analytical Evaluation

Incomplete: 10/11 questions answered. Please provide detailed responses for all questions.



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